

Steps to Design

- 1) Decide the number of Flip Flops
- 2) Excitation table of Flip Flop
- 3) State diagrams and Excitation Circuit
- 4) obtain simplified equation using K map
- 5) Draw the logic Diagram.

2 Bit using Jk Flip Flops.

Q_1	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Q_1	Q_2	Q_1^+	Q_2^+	J_1	K_1	J_2	K_2
0	0	0	1	0	x	1	x
0	1	1	0	1	x	x	1
1	0	1	1	x	0	1	x
1	1	0	0	x	1	x	1

Q_1	Q_2	J_1
0	0	0
0	1	1
1	0	x
1	1	1

$$J_1 = Q_2$$

Q_1	Q_2	K_1
0	0	x
0	1	x
1	0	1
1	1	0

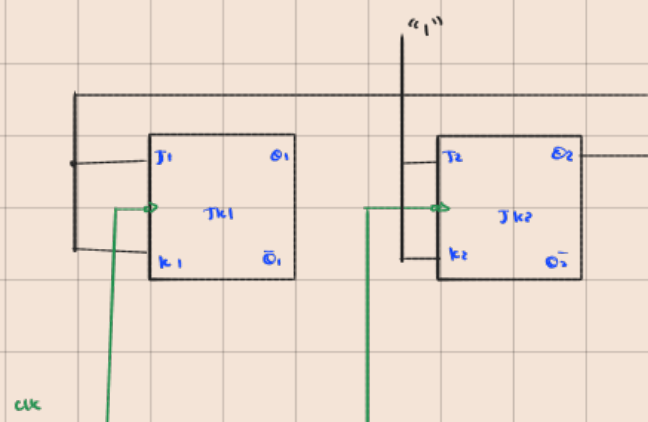
$$K_1 = Q_2$$

Q_1	Q_2	J_2
0	0	1
0	1	x
1	0	1
1	1	x

$$J_2 = 1$$

Q_1	Q_2	K_2
0	0	x
0	1	1
1	0	x
1	1	1

$$K_2 = 1$$



3-bit Up counter.

Q_1	Q_2	J	K	Q_1	Q_2	Q_3	Q_1	Q_2	Q_3	J_1	K_1	J_2	K_2	J_3	K_3
0	0	0	x	0	0	0	0	0	1	0	x	0	x	1	x
0	1	1	x	0	0	1	0	1	0	0	x	1	x	x	1
1	0	x	1	0	1	0	0	1	1	0	x	x	0	1	x
1	1	x	0	0	1	1	1	0	0	1	x	x	1	x	1
				1	0	0	1	0	1	x	0	0	x	1	x
				1	0	1	1	1	0	x	0	1	x	x	1
				1	1	0	1	1	1	x	0	x	0	1	x
				1	1	1	0	0	0	x	1	x	1	x	1

J_1

$Q_1 Q_3$	00	01	11	10
Q_1	0	0	1	0
0	0	0	1	0
1	x	x	x	x

$$J_1 = Q_2 Q_3$$

J_2

$Q_1 Q_3$	00	01	11	10
Q_1	0	1	x	x
0	0	1	x	x
1	0	1	x	x

$$J_2 = Q_3$$

J_3

$Q_1 Q_3$	00	01	11	10
Q_1	1	x	x	1
0	1	x	x	1
1	1	x	x	1

$$J_3 = 1$$

K_1

$Q_1 Q_3$	00	01	11	10
Q_1	x	x	1	x
0	x	x	1	x
1	0	0	1	0

$$K_1 = Q_2 Q_3$$

K_2

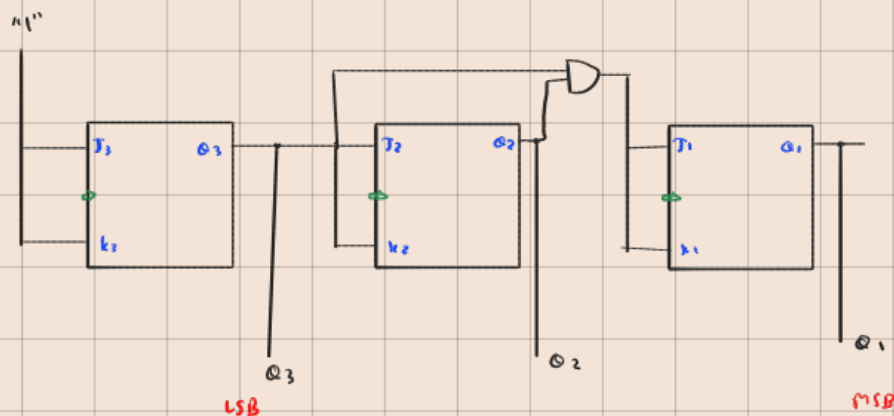
$Q_1 Q_3$	00	01	11	10
Q_1	x	x	1	0
0	x	x	1	0
1	x	x	1	0

$$K_2 = Q_3$$

K_3

$Q_1 Q_3$	00	01	11	10
Q_1	x	1	1	x
0	x	1	1	x
1	x	1	1	x

$$K_3 = 1$$



2 Bit using T flip flops:

Q	$Q+1$	T
0	0	0
0	1	1
1	0	1
1	1	0

Q_1	Q_2	Q_3^1	Q_1^1	Q_2^1	Q_3^1	T_1	T_2	T_3
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

T_1

$Q_1 Q_3$	00	01	11	10
0	0	0	1	0
1	0	0	1	0

$$T_1 = Q_2 Q_3$$

T_2

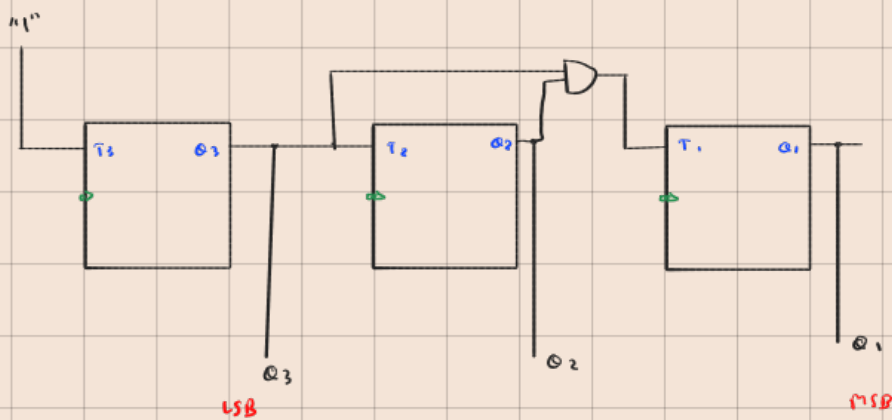
$Q_1 Q_3$	00	01	11	10
0	0	1	1	0
1	0	1	1	0

$$T_2 = Q_3$$

T_3

$Q_1 Q_3$	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$$T_3 = 1$$



2 Bit using 0 flip flops:

Q	Q_{n+1}	P	Q_1	Q_2	Q_3	Q_1^1	Q_2^1	Q_3^1	Q_1	Q_2	Q_3
0	0	0	0	0	0	0	0	1	0	0	1
0	1	1	0	0	1	0	1	0	0	1	0
1	0	0	1	1	0	0	1	1	0	1	1
1	1	1	0	1	1	1	0	0	1	0	0
			1	0	0	1	0	1	1	0	1
			1	0	1	1	1	0	1	1	0
			1	1	0	1	1	1	1	1	1
			1	1	1	0	0	0	0	0	0

D_1

$Q_1 Q_2$	00	01	11	10
Q_1	0	0	1	0
Q_2	1	1	0	1

$$D_1 = Q_1 \bar{Q}_2 + \bar{Q}_1 Q_2 Q_3 + Q_1 Q_2 \bar{Q}_3$$

$$= Q_1 (\bar{Q}_2 + Q_2 \bar{Q}_3) + \bar{Q}_1 Q_2 Q_3$$

D_2

$Q_1 Q_2$	00	01	11	10
Q_1	0	1	0	1
Q_2	0	1	0	1

$$D_2 = \bar{Q}_2 Q_3 + Q_2 \bar{Q}_3$$

$$D_2 = (Q_2 \oplus Q_3)$$

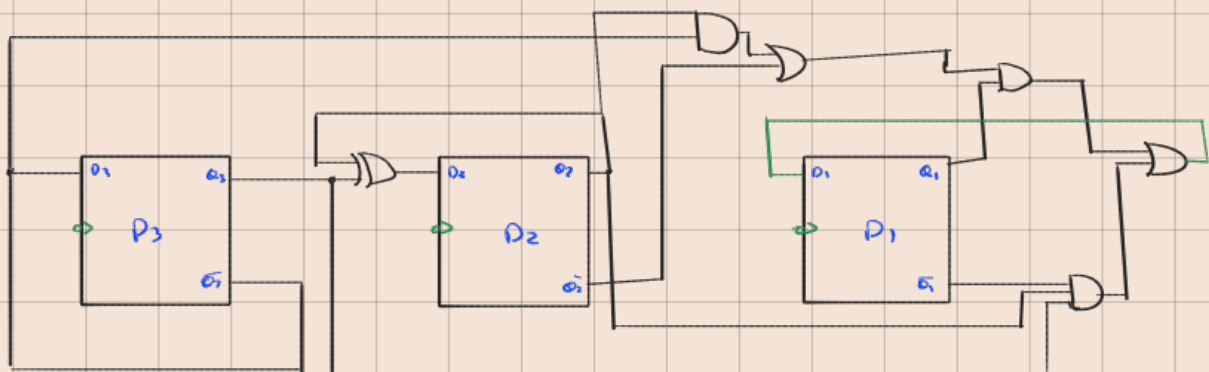
D_3

$Q_1 Q_2$	00	01	11	10
Q_1	1	0	0	1
Q_2	1	0	0	1

$$D_3 = \bar{Q}_1 \bar{Q}_3 + Q_2 \bar{Q}_3$$

$$= \bar{Q}_3 (\bar{Q}_1 + Q_2)$$

$$= \bar{Q}_3$$



4 bit Decade Counter using JK

Q₁ Q₂ Q₃ Q₄

0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1

1

10 → Reset condition

excitation table

Q	Q _{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Q₁ Q₂ Q₃ Q₄ Q₁⁺ Q₂⁺ Q₃⁺ Q₄⁺ J

0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0

mod 5 counter using T Flip Flops synchronous

Q	Q ₂	T		Q ₁	Q ₂	Q ₃	Q ₁ ⁺	Q ₂ ⁺	Q ₃ ⁺	T ₁	T ₂	T ₃
0	0	0	0	0	0	0	0	0	1	0	0	1
1	1	1	1	0	0	1	0	1	0	0	1	1
2	0	1	2	1	1	0	0	1	1	0	0	1
3	1	0	3	0	1	1	1	0	0	1	1	1
4	1	0	4	1	0	0	1	0	1	0	0	1
5	1	0	5	1	0	1	0	0	0	1	0	1

T_1

Q ₁ Q ₃	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$T_1 = Q_2 Q_3 + Q_1 Q_3$$

$$= Q_3 (Q_2 + Q_1)$$

T_2

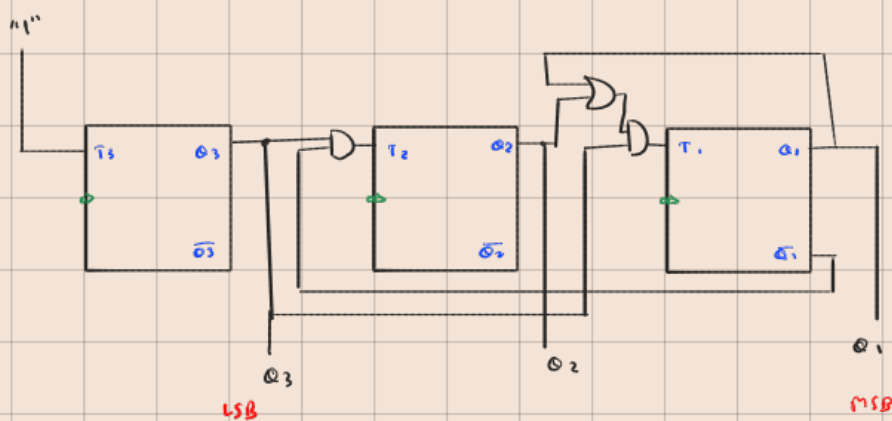
Q ₁ Q ₃	00	01	11	10
0	0	1	1	0
1	0	0	1	1

$$T_2 = \bar{Q}_1 Q_3$$

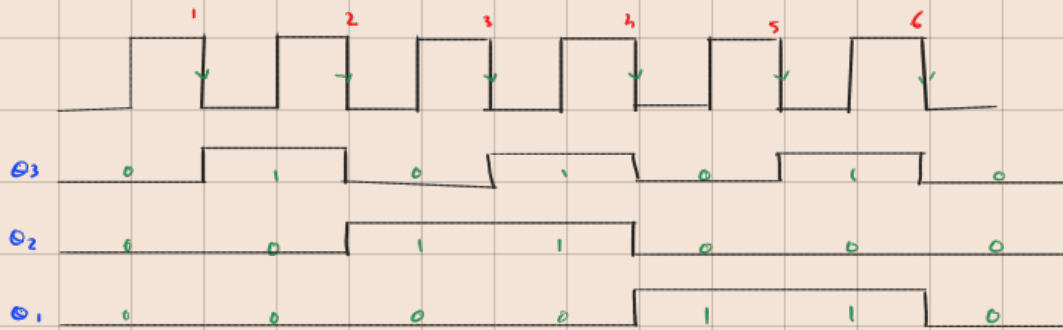
T_3

Q ₁ Q ₃	00	01	11	10
0	1	1	1	1
1	1	1	1	1

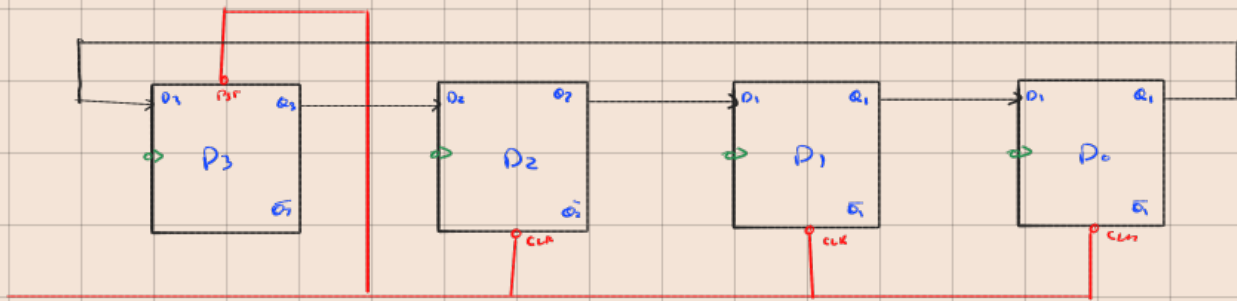
$$T_3 = 1$$



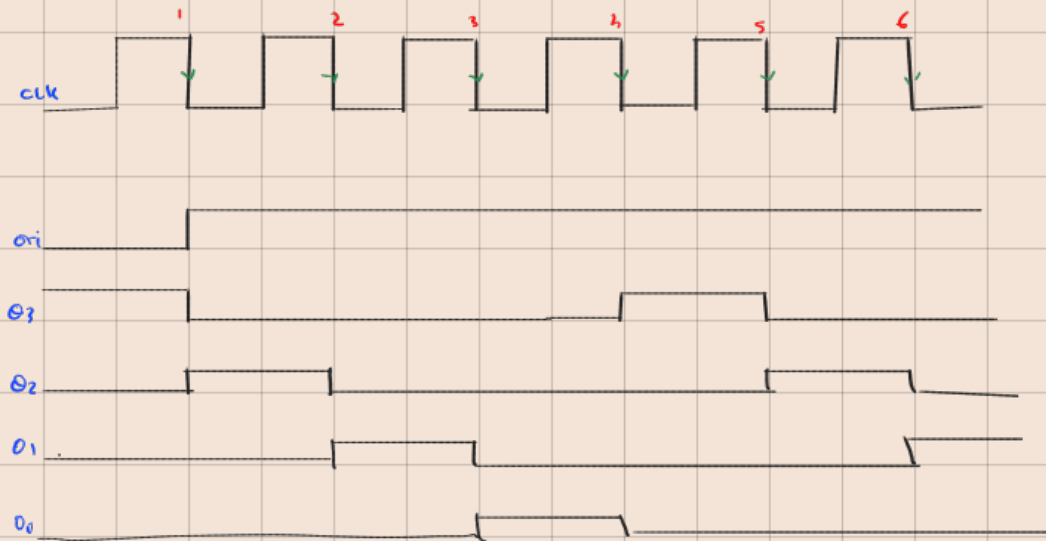
row 6 wave form

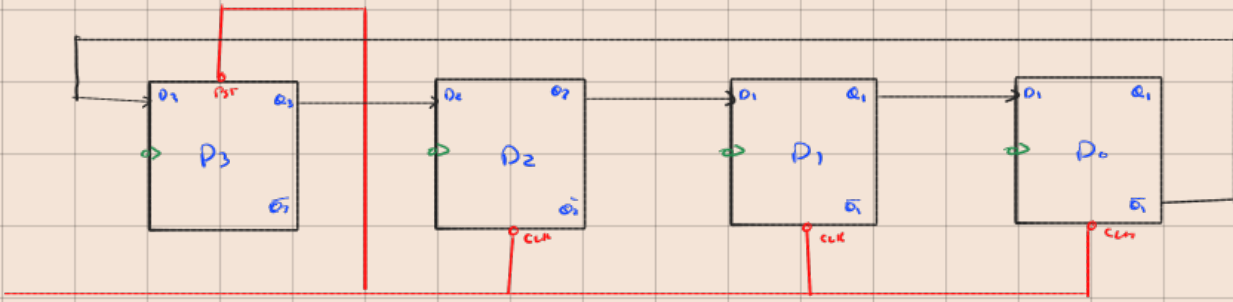


Ring Counter



ori	clk	Q_3	Q_2	Q_1	Q_0
0	x	1	0	0	0
1		0	1	0	0
1		0	0	1	0
1		0	0	0	1
1		1	0	0	0





CLK	CLK	Q_3	Q_2	Q_1	Q_0
0	x	1	0	0	0
1		1	1	0	0
1		1	1	1	0
1		1	1	1	1
1		0	1	1	1
		0	0	1	1
		0	0	0	1
		0	0	0	0

